

Student
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Total Points
86 / 100 pts

Question 1

(no title)

15 / 15 pts

✓ - 0 pts Correct

Question 2

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 1 pt Mass flow rate should be around 4.9 kg/s.
- 1 pt Area should be around 0.07 sq. m

Question 3

(no title)

14 / 15 pts

- 0 pts Correct

✓ - 1 pt Power should be around 3500 kW.

Question 4

(no title)

12 / 15 pts

- 0 pts Correct

✓ - 3 pts Incomplete

- 1 pt Heat transfer rate will be around -15 kW.

Question 5

(no title)

16 / 20 pts

- 0 pts Correct

- 1 pt POWER=0.52kW

✓ - 4 pts Incomplete

Question 6

(no title)

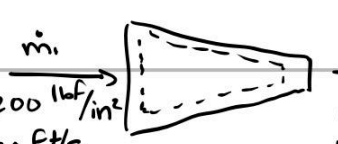
14 / 20 pts

- 0 pts Correct

- 1 pt x=0.191

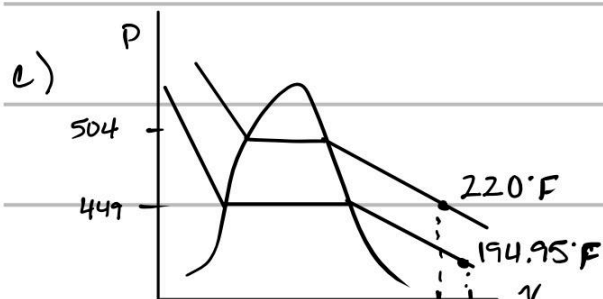
✓ - 6 pts incomplete

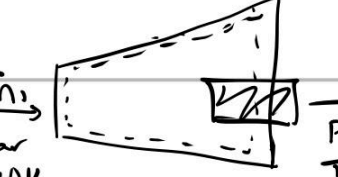
No questions assigned to the following page.

1)  $\dot{Q} = 0$ $\dot{W} = 0$ b) steady state, $\dot{m}_1 = \dot{m}_2$, $\dot{Q} = \dot{W} = 0$
adiabatic, $\Delta PE = 0$

c) $\dot{m} = \rho_1 A_1 V_1 = \rho_2 A_2 V_2$ $h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2}$

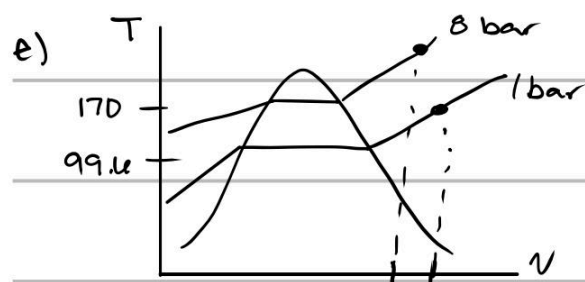
d) $h_1 = 144.2 \text{ BTU/lb}$ $h_2 = h_1 = 144.2 \text{ BTU/lb}$ so $T_2 \Rightarrow \frac{145.23 - 140.68}{200 - 180} = \frac{144.2 - 140.8}{T_2 - 180} \rightarrow T_2 = 194.95 \text{ } ^\circ\text{F}$



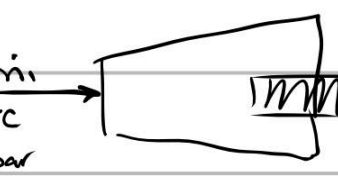
2)  b) S.S, $\dot{m}_1 = \dot{m}_2$, $\Delta PE = 0$, ideal gas, $\dot{W} = 2500 \text{ kW}$

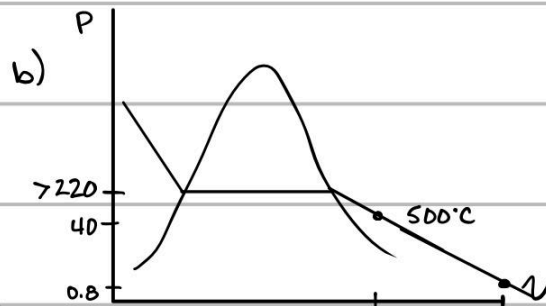
c) $\dot{m} (h_1 + \frac{V_1^2}{2}) = \dot{m} (h_2 + \frac{V_2^2}{2}) + \dot{W} \rightarrow \dot{m} = \frac{\dot{W}}{h_1 - h_2 - \frac{V_2^2}{2}}$

d) $\dot{m} = \frac{\dot{W}}{c_p(T_1 - T_2) - \frac{V_2^2}{2}} = \frac{2500}{1.005(900 - 450) - \frac{90^2}{2}} = 4.93 \text{ kg/s}$



$A_c = \frac{4.93(287)(450)}{100000(90)} = 0.071 \text{ m}^2$

3) 

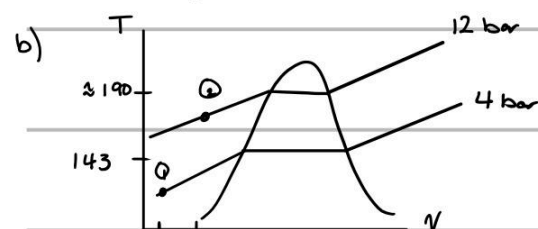


c) steady state, adiabatic, $\Delta PE = 0$ $v_{g2} \Rightarrow \frac{5.095 - 3.565}{1 - 0.7} = \frac{x - 5.095}{0.8 - 0.7} \rightarrow v_{g2} = 4.585 \text{ @ } 500^\circ\text{C}$

d) $\dot{m} = \frac{\dot{V}_2}{v_{g2}} = \frac{9.48}{4.585} = 2.07 \text{ kg/s}$

$\dot{W} = \dot{m} \left((h_1 - h_{g2}) + \frac{V_1^2 - V_2^2}{2} \right) \rightarrow \dot{W} = 2.07 \left((802) + \frac{40000 - 22500}{2(1000)} \right) = 1678.25 \text{ kW}$

4) a)  $\dot{W}_{in} = \dot{m} \cdot 60 \text{ kJ/kg}$



c) steady state, ideal fluid, $\Delta KE = \Delta PE = 0$

d) $\dot{m} = \frac{\dot{V}_1}{v_1}$ $\dot{m} h_1 + \dot{W}_{in} + \dot{Q} = \dot{m} h_2 \rightarrow \dot{Q} = \dot{m} (h_2 - h_1) - \dot{W}_{in}$

$\dot{m} = \frac{0.01667}{0.05006} = 1.32 \text{ kg/s}$

$\dot{Q} = 1.32 (1285.8 - 2626) - 60 = -35.4 \text{ kW}$